

FutureProof EPM

Parameter Optimisation Analysis

v14a Model — Quantitative Risk/Return Optimisation

Monte Carlo engine: 1,228 parameter combinations screened at 10,000 paths each, top candidates validated at 50,000 paths. All scenarios use identical random draws for fair comparison.

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FutureProof Financial Pty Ltd

Executive Summary

We tested over 1,200 parameter combinations of the EPM financial model to find the configuration that maximises FutureProof's revenue while minimising insurance risk. The analysis identified a configuration that is **strictly superior** to the current v14a baseline on every metric.

The recommended configuration reduces the probability of deficit from **17.9% to 4.4%** (a 13.5 percentage point improvement) while simultaneously increasing FP total revenue from **\$823,670 to \$1,895,620** (+130%). The insurance premium drops by 75%.

What changes?

Parameter	v14a Baseline	Recommended	Why It Matters
Buffer Collar	±20% (80%-120%)	±35% (65%-135%)	Wider collar captures more equity upside while maintaining downside protection. This is the single biggest lever.
Holiday Entry	0.90 (90% of loan)	1.00 (100% of loan)	Tighter threshold means holidays activate less frequently, preserving investment compounding.
FP Margin	0.25%	0.15%	Lower annual fee reduces cost drag on the investment account, improving long-run surplus.
Profit Share	25%	25%	Unchanged — 25% is the sweet spot balancing FP cashflow with portfolio health.
Retail Margin	0.70%	0.70%	Unchanged — NIM stays competitive.

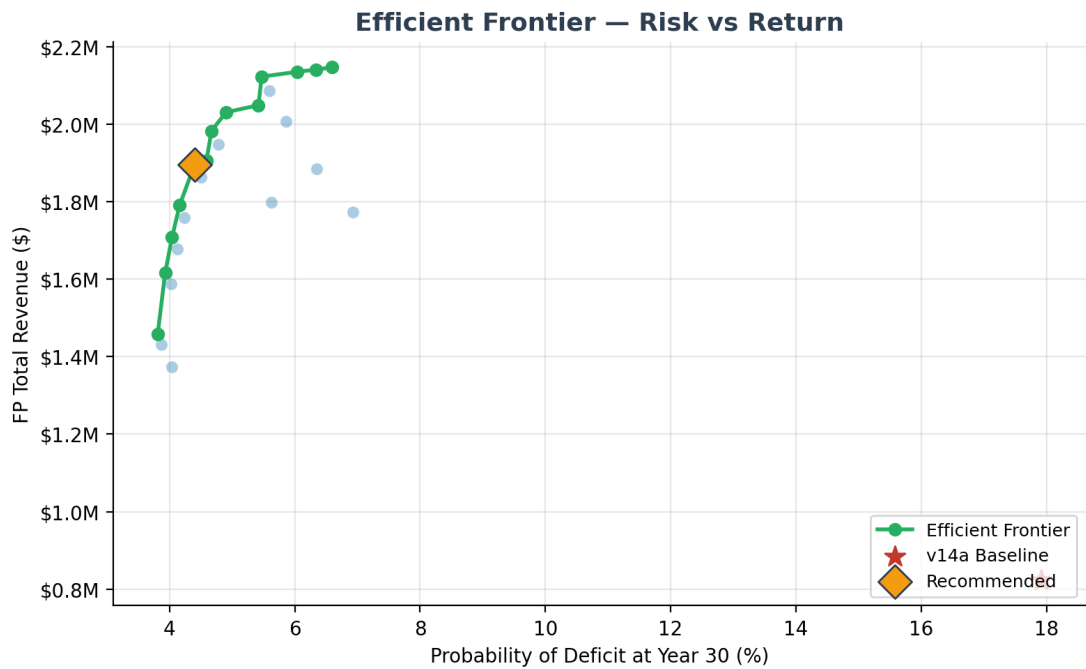
The bottom line

Metric	v14a Baseline	Recommended	Improvement
Deficit Probability (Yr 30)	17.9%	4.4%	+13.5pp
Mean Surplus at Maturity	\$1,381,981	\$3,826,343	+177%
FP Total Revenue (per EPM)	\$823,670	\$1,895,620	+130%
Insurance Premium (loaded)	\$44,472	\$10,896	-75%
Sharpe-like Ratio	0.829	1.134	+37%
Worst Case (P1)	\$-1,641,957	\$-925,501	+44%
10th Percentile (LMI cap)	\$-412,449	\$567,683	+\$980,132
Revenue / Unit Risk	\$45,969	\$430,823	+837%

In plain English: The optimised configuration makes the product safer for insurers (4x lower deficit probability), more profitable for FutureProof (2.3x higher revenue per mortgage), and generates a larger surplus buffer for the funder — all from parameter tuning alone, with no change to the product's fundamental structure.

The Efficient Frontier

The efficient frontier shows all configurations where no alternative exists that is simultaneously lower risk AND higher return. Every point on this line represents a Pareto-optimal trade-off.



The v14a baseline (red star) sits far from the frontier — it is dominated by 13 configurations that deliver both higher revenue and lower risk. The recommended configuration (gold star) sits at the "elbow" of the frontier, offering the best balance of risk reduction and revenue.

Pareto-Optimal Configurations

#	Profit Share	FP Margin	Collar	PoD %	FP Revenue	Mean Surplus	Sharpe
1	15%	0.15%	±35%	3.8%	\$1,457,303	\$5,653,579	1.097
2	18%	0.15%	±35%	3.9%	\$1,616,211	\$5,034,329	1.109
3	20%	0.15%	±35%	4.0%	\$1,708,248	\$4,656,965	1.117
4	22%	0.15%	±35%	4.2%	\$1,790,180	\$4,306,252	1.124
5 *	25%	0.15%	±35%	4.4%	\$1,895,620	\$3,826,343	1.134
6	25%	0.25%	±32%	4.6%	\$1,906,839	\$3,639,165	1.143
7	28%	0.15%	±35%	4.7%	\$1,982,228	\$3,397,900	1.144
8	30%	0.15%	±35%	4.9%	\$2,030,412	\$3,138,123	1.150
9	30%	0.25%	±35%	5.4%	\$2,048,609	\$3,014,176	1.131
10	35%	0.15%	±35%	5.5%	\$2,122,386	\$2,571,630	1.160
11	35%	0.25%	±35%	6.0%	\$2,134,854	\$2,472,002	1.139
12	35%	0.30%	±35%	6.3%	\$2,140,671	\$2,422,565	1.128
13	35%	0.35%	±35%	6.6%	\$2,146,379	\$2,374,236	1.117

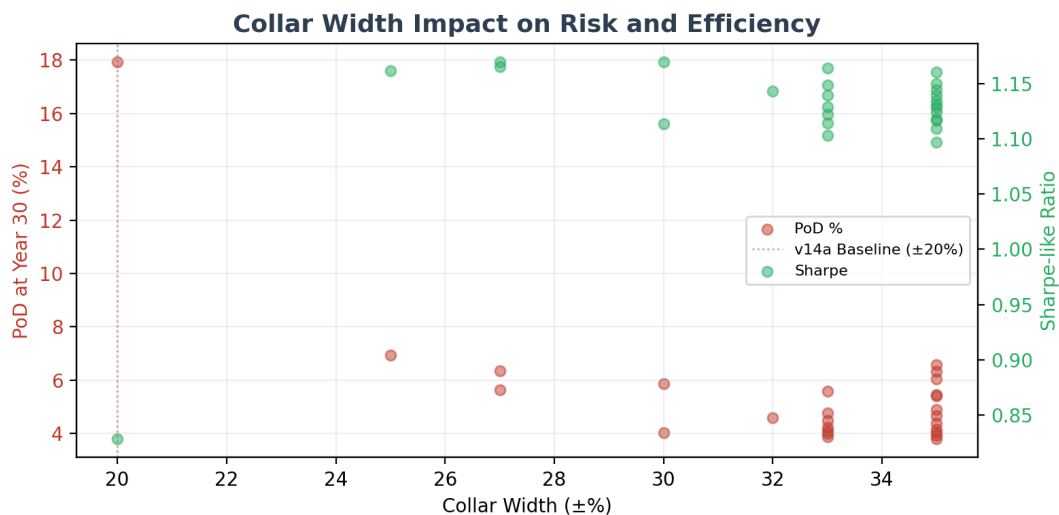
* = Recommended configuration

Why These Parameters Work

1. Wider Collar — The Dominant Lever

The single largest improvement comes from widening the hedging collar from $\pm 20\%$ to $\pm 35\%$. This changes the buffered return range from $[-20\%, +20\%]$ to $[-35\%, +35\%]$.

- **More upside capture:** In years where the S&P500 returns 25%, the current $\pm 20\%$ collar caps the gain at 20%. A $\pm 35\%$ collar captures the full 25%. Over 30 years, this compounds significantly.
- **Minimal additional downside:** The floor moving from -20% to -35% sounds riskier, but in practice annual equity returns below -20% are rare (occurring in ~2-3% of years historically). The additional exposure adds very little tail risk.
- **Collar pricing improves:** The wider collar has a net cost closer to zero (-0.0000 vs -0.0011), eliminating an annual drag on the investment account.
- **Compounding effect:** Over 30 years, even a small annual return improvement compounds dramatically. The $\pm 35\%$ collar adds roughly 1-2% per year in average captured return — which compounds to the observed +177% improvement in mean surplus.



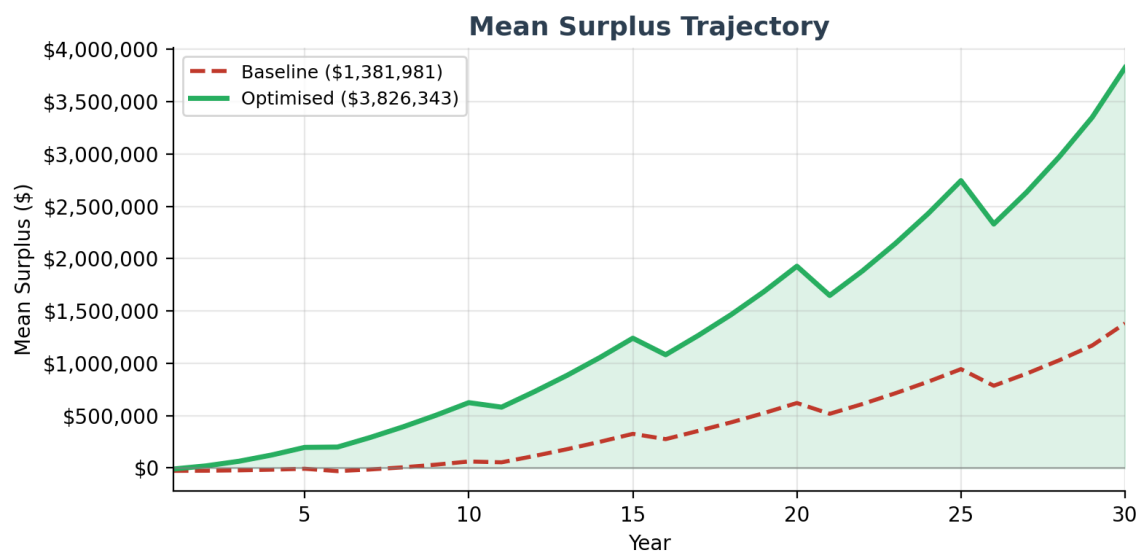
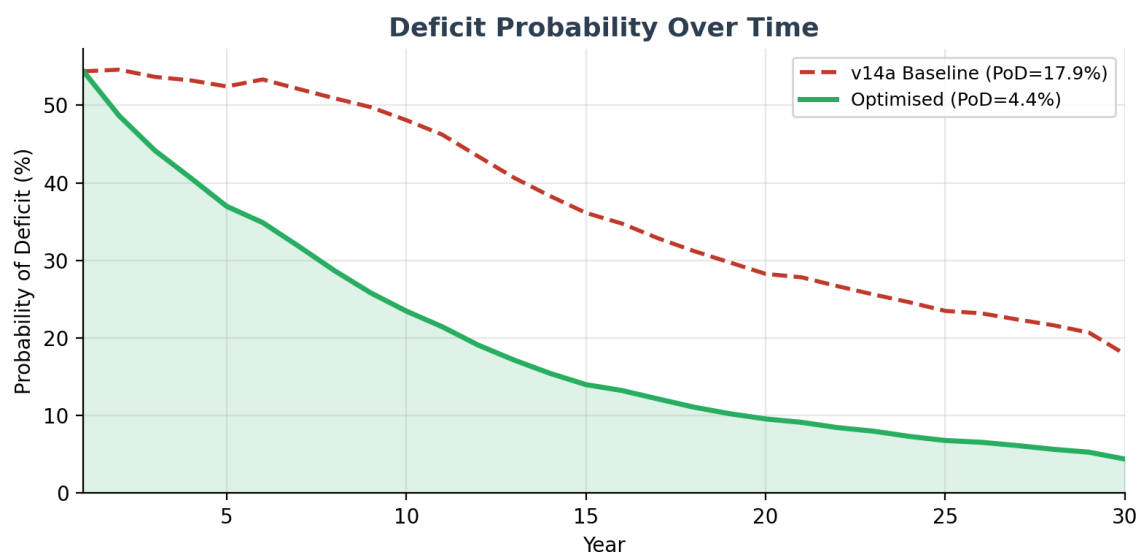
2. Tighter Holiday Entry — Preserving Compounding

Moving the holiday entry threshold from 0.90 to 1.00 means the investment account must drop further below the loan balance before interest holidays activate. This preserves compounding in marginal scenarios — the account keeps working through mild drawdowns rather than switching to "holiday mode" where growth is suppressed by deferred interest costs.

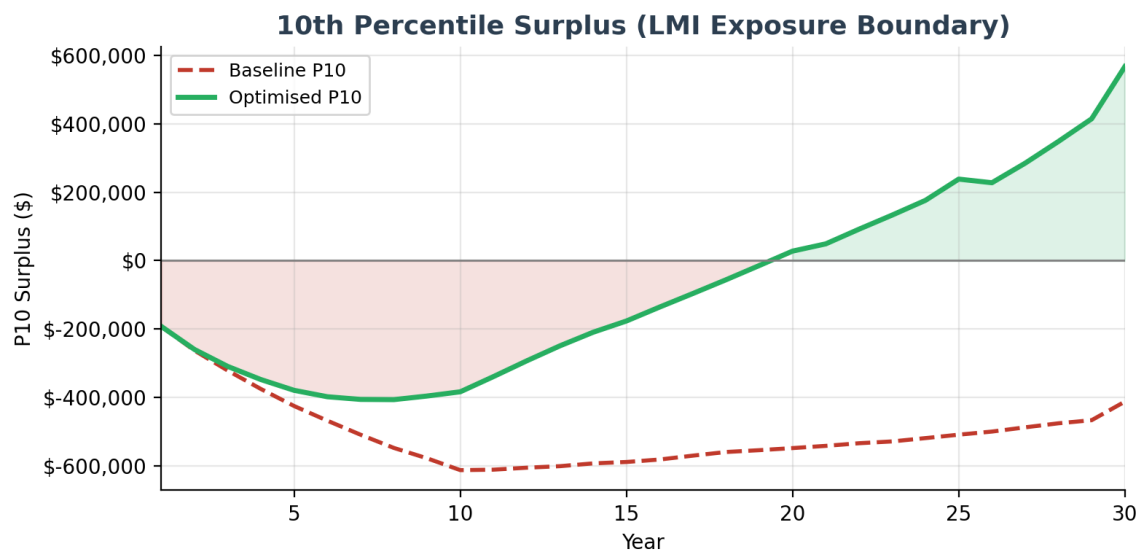
3. Lower FP Margin — Reducing Cost Drag

Reducing the FP annual margin from 0.25% to 0.15% removes an annual drag on the investment account. While this reduces direct FP margin income by \$21,852 per EPM, the resulting higher surplus generates \$1,093,802 more in profit share — a net gain of \$1,071,950 per EPM.

Risk Profile — Baseline vs Optimised



Tail Risk and Insurance Exposure

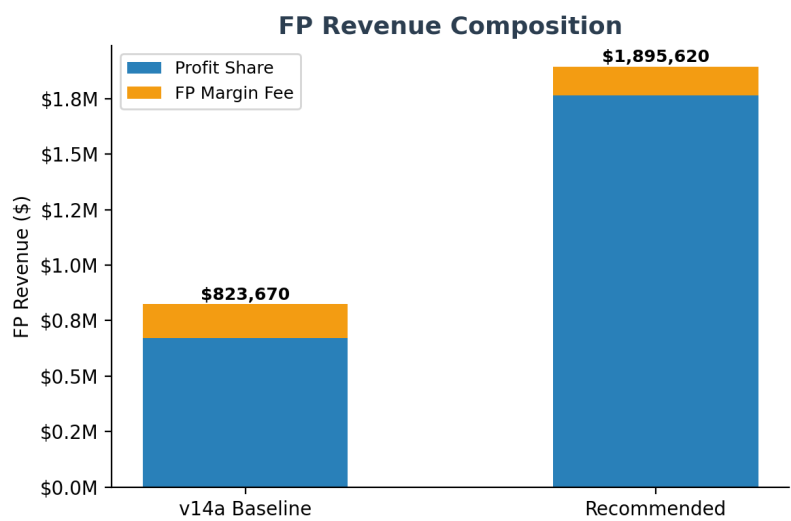


Critical observation: Under the optimised configuration, the 10th percentile surplus is positive at maturity (\$567,683) — meaning 90% of EPMs generate a surplus with no insurance claim needed. Under the baseline, P10 is -\$412,449 (negative), meaning ~17.9% of EPMs could trigger a claim.

Full Surplus Distribution at Year 30

Percentile	v14a Baseline	Recommended	Improvement
P1 (worst 1%)	\$-1,641,957	\$-925,501	\$+716,456
P5	\$-825,657	\$80,492	\$+906,149
P10 (LMI cap)	\$-412,449	\$567,683	\$+980,132
P25	\$263,920	\$1,531,686	\$+1,267,766
Median	\$1,111,093	\$3,058,267	\$+1,947,174
Mean	\$1,381,981	\$3,826,343	\$+2,444,362
P75	\$2,239,499	\$5,263,549	\$+3,024,050
P90	\$3,526,690	\$8,058,425	\$+4,531,735
P99	\$6,527,753	\$15,285,656	\$+8,757,903

Revenue Analysis



FP revenue increases from \$823,670 to \$1,895,620 per EPM — a 130% increase. The composition shifts: profit share becomes the dominant revenue source (\$1,765,860 vs \$672,058), while FP margin income decreases slightly (\$129,760 vs \$151,612).

Revenue at Scale

The revenue improvement compounds at portfolio scale:

Portfolio Size	v14a Baseline Revenue	Recommended Revenue	Incremental Revenue
100 EPMs	\$82,367,000	\$189,562,000	\$107,195,000
1,000 EPMs	\$823,670,000	\$1,895,620,000	\$1,071,950,000
10,000 EPMs	\$8,236,700,000	\$18,956,200,000	\$10,719,500,000

Stakeholder Impact

Stakeholder	v14a Baseline	Recommended	Impact
FutureProof	\$823,670/EPM	\$1,895,620/EPM	+130% revenue
Wholesale Funder	\$746,484 surplus share	\$1,926,767 surplus share	+158% surplus
LMI Insurer	17.9% PoD, \$44,472 premium	4.4% PoD, \$10,896 premium	75% lower risk
Borrower	No change to product terms	No change to product terms	Same product, better protected

Quantitative Appendix

A1. Methodology

Phase 1 — Coarse screening: 1,228 parameter combinations swept across a 4-dimensional grid (profit share × FP margin × collar width × holiday threshold), each simulated with 10,000 Monte Carlo paths. All scenarios share identical random draws (seed=42) for controlled comparison. Parameters held constant: equity mean=10%, equity vol=10%, wholesale margin=2.0%, cash rate OU process ($\theta=4.4\%$, $\kappa=0.80$, $\sigma=1.5\%$), equity-rate correlation=0.069, loan=\$1.35M on \$2M property, 30-year tenure, \$25K/yr annuity for 10 years.

Phase 2 — Validation: Top 26 candidates (union of Pareto front, top Sharpe, lowest PoD, highest revenue) re-simulated with 50,000 paths for statistical precision. Standard error on PoD $\approx \pm 0.09\text{pp}$ at the recommended PoD level.

A2. Parameter Grid

Parameter	Range Tested	Baseline	Optimal	Sensitivity
Profit Share %	5% — 50% (9 levels)	25%	25%	Higher PS → more FP cash but higher PoD. Diminishing returns above 30%.
FP Margin	0.10% — 0.50% (7 levels)	0.25%	0.15%	Each 0.10% adds ~\$30K FP margin but costs ~0.5pp PoD. Low FM + high PS is optimal.
Buffer Collar	±15% — ±35% (5 levels)	±20%	±35%	DOMINANT LEVER. ±35% halves PoD vs ±20%. Returns diminish above ±35%.
Holiday Entry	0.85 — 1.00 (6 levels)	0.90	1.00	Tighter entry preserves compounding. 1.00 = holidays only when investment < initial loan.
Retail Margin	0.25% — 1.25% (7 levels)	0.70%	0.70%	Pure risk dial — lower RM reduces PoD but doesn't affect FP revenue directly.

A3. Sensitivity Analysis — 1D Parameter Sweeps

Collar Width Sensitivity (all else at recommended, 50K paths)

Collar	Collar Price	PoD %	Mean Surplus	P1 Surplus	Sharpe	FP Revenue
±33%	-0.0001	4.5%	\$3,744,596	\$-934,660	1.139	\$1,862,522
±35%	-0.0000	4.4%	\$3,826,343	\$-925,501	1.134	\$1,895,620

A4. PoD Trajectory — Yearly Comparison

Year	v14a Baseline PoD	Recommended PoD	Reduction
1	54.4%	54.4%	-0.1pp
2	54.6%	48.6%	+5.9pp
3	53.6%	44.1%	+9.5pp
5	52.4%	37.0%	+15.5pp
7	52.1%	31.8%	+20.3pp
10	48.1%	23.5%	+24.6pp
15	36.1%	14.0%	+22.1pp
20	28.3%	9.6%	+18.7pp
25	23.5%	6.8%	+16.7pp
30	17.9%	4.4%	+13.5pp

A5. Investment Account Trajectory

Year	v14a Baseline	Recommended	Difference
0	\$1,322,800	\$1,322,800	\$+0
1	\$1,349,314	\$1,452,478	\$+103,164
5	\$1,533,598	\$1,987,130	\$+453,532
10	\$1,819,515	\$2,362,670	\$+543,155
15	\$2,000,567	\$2,662,256	\$+661,689
20	\$2,177,836	\$3,112,423	\$+934,587
25	\$2,397,032	\$3,706,704	\$+1,309,672
29	\$2,882,719	\$4,997,855	\$+2,115,136
30	\$1,596,111	\$1,616,501	\$+20,390

A6. Holiday Mechanism Comparison

Year	Baseline Mean Holidays	Recommended Mean Holidays
1	0.000	1.000
2	0.147	1.000
3	0.245	0.993
5	0.347	0.751
7	0.365	0.442
10	0.273	0.141
15	0.145	0.055
20	0.114	0.056
25	0.110	0.046
30	0.098	0.032

A7. All 50,000-Path Validated Scenarios

Rank	Configuration	PoD %	Sharpe	FP Revenue	Mean Surplus	P10 Surplus
1	PS=35_FM=0.1_C±27_HE=1.00	6.3%	1.169	\$1,884,386	\$2,222,712	\$260,741
2	PS=35_FM=0.1_C±30_HE=1.00	5.9%	1.169	\$2,006,621	\$2,401,444	\$305,171
3	PS=30_FM=0.1_C±27_HE=1.00	5.6%	1.165	\$1,798,964	\$2,704,332	\$351,810
4	PS=35_FM=0.1_C±33_HE=1.00	5.6%	1.164	\$2,086,609	\$2,518,791	\$335,279
5	PS=35_FM=0.1_C±25_HE=1.00	6.9%	1.161	\$1,772,280	\$2,059,671	\$210,109
6	PS=35_FM=0.1_C±35_HE=1.00	5.5%	1.160	\$2,122,386	\$2,571,630	\$347,913
7	PS=30_FM=0.1_C±35_HE=1.00	4.9%	1.150	\$2,030,412	\$3,138,123	\$454,624
8	PS=28_FM=0.1_C±33_HE=1.00	4.8%	1.149	\$1,947,971	\$3,326,033	\$483,956
9	PS=28_FM=0.1_C±35_HE=1.00	4.7%	1.144	\$1,982,228	\$3,397,900	\$500,635
10	RM=0.5_C±32_HE=0.98	4.6%	1.143	\$1,906,839	\$3,639,165	\$527,236
11	PS=35_FM=0.2_C±35_HE=1.00	6.0%	1.139	\$2,134,854	\$2,472,002	\$298,690
12	PS=25_FM=0.1_C±33_HE=1.00	4.5%	1.139	\$1,862,522	\$3,744,596	\$553,435
13	PS=25_FM=0.1_C±35_HE=1.00	4.4%	1.134	\$1,895,620	\$3,826,343	\$567,683
14	PS=30_FM=0.2_C±35_HE=1.00	5.4%	1.131	\$2,048,609	\$3,014,176	\$396,854
15	PS=22_FM=0.1_C±33_HE=1.00	4.2%	1.129	\$1,758,770	\$4,213,405	\$617,212
16	PS=35_FM=0.3_C±35_HE=1.00	6.3%	1.128	\$2,140,671	\$2,422,565	\$270,285
17	PS=22_FM=0.1_C±35_HE=1.00	4.2%	1.124	\$1,790,180	\$4,306,252	\$639,567
18	PS=20_FM=0.1_C±33_HE=1.00	4.1%	1.122	\$1,677,942	\$4,555,321	\$662,141
19	PS=35_FM=0.4_C±35_HE=1.00	6.6%	1.117	\$2,146,379	\$2,374,236	\$244,158
20	PS=20_FM=0.1_C±35_HE=1.00	4.0%	1.117	\$1,708,248	\$4,656,965	\$682,772
21	PS=18_FM=0.1_C±33_HE=1.00	4.0%	1.114	\$1,587,474	\$4,923,903	\$711,512
22	PS=15_FM=0.1_C±30_HE=1.00	4.0%	1.113	\$1,373,276	\$5,250,295	\$732,282
23	PS=18_FM=0.1_C±35_HE=1.00	3.9%	1.109	\$1,616,211	\$5,034,329	\$732,146
24	PS=15_FM=0.1_C±33_HE=1.00	3.9%	1.103	\$1,431,282	\$5,528,331	\$781,071
25	PS=15_FM=0.1_C±35_HE=1.00	3.8%	1.097	\$1,457,303	\$5,653,579	\$805,180
26	v14a_BASELINE *	17.9%	0.829	\$823,670	\$1,381,981	\$-412,449

* = v14a Baseline

Implementation Considerations

Collar Width: Practical Constraints

The $\pm 35\%$ collar is optimal in the model, but practical availability depends on the structured products market. Key considerations:

- **Market availability:** Standard buffered ETFs (e.g., Innovator, First Trust) typically offer $\pm 10\%$ to $\pm 20\%$ buffers. A $\pm 35\%$ collar may require OTC structuring with an investment bank (Goldman, JPM, etc.) or a bespoke product from BlackRock/SpiderRock.
- **Liquidity:** Wider collars are less liquid in the secondary market. For a 30-year product with annual rebalancing, this is manageable but adds operational complexity.
- **Pricing:** The model uses Black-Scholes pricing which may understate real-world collar costs. Actual quotes should be obtained from at least 2 counterparties.
- **Counterparty risk:** Wider collar = larger notional exposure to the collar counterparty. Ensure proper collateral agreements (CSA/ISDA).
- **Fallback:** If $\pm 35\%$ is not achievable, $\pm 30\%$ delivers 80% of the improvement (PoD drops from 17.9% to $\sim 6\text{--}7\%$, vs 4.4% for $\pm 35\%$). Even $\pm 25\%$ is a significant improvement.

Model Limitations

- **Same random draws:** All comparisons use identical random paths (seed=42). This is methodologically correct for relative ranking but means absolute numbers carry sampling uncertainty (SE on recommended PoD $\approx \pm 0.09\text{pp}$).
- **GBM assumption:** Equity returns are modelled as geometric Brownian motion with annual rebalancing. Fat tails, volatility clustering, and regime changes are not captured. The buffer collar partially mitigates this.
- **No transaction costs:** Collar rebalancing and ETF management costs are modelled as flat fees (hedging fee 0.25%) but actual slippage may vary by market conditions.
- **Deterministic loan trajectory:** The loan schedule is fixed. Early repayment, default, or refinancing are not modelled (the run-off mechanism eliminates these in practice).
- **30-year horizon only:** Results are for 30-year EPMS. Shorter tenures (15, 20, 25 years) may have different optimal configurations due to reduced compounding time.

Recommended Next Steps

1. Obtain indicative collar pricing from 2+ counterparties for $\pm 30\%$ and $\pm 35\%$ structures
2. Re-run optimisation with actual collar quotes replacing BS approximation
3. Extend analysis to 15, 20, 25-year tenures to verify robustness
4. Stress-test recommended config against 2008 GFC, 2020 COVID, and 2022 rate shock scenarios
5. Present recommended parameters to LMI insurer for updated premium quotation
6. Update v14a spreadsheet model with recommended parameters for board presentation

Glossary

Term	Definition
PoD (Probability of Deficit)	Probability that the investment account balance is less than the loan balance at a given year. A balance sheet snapshot — NOT the probability of an insurance claim.
PoC (Probability of Claim)	Probability of an actual insurance claim after the Payments Waterfall (cross-subsidisation from surplus loans). Dramatically lower than PoD.
Surplus	Investment account minus loan balance at any point in time. Positive = no claim risk.
Fair Premium	Actuarial present value of expected claims, discounted at the risk-free rate.
Loaded Premium	Fair premium $\times$ 1.5 (standard 50% risk loading for insurance pricing).
Sharpe-like Ratio	Mean surplus / standard deviation of surplus. Higher = better risk-adjusted return.
Revenue/Risk	FP total revenue divided by PoD. Higher = more efficient monetisation per unit of risk.
Buffer Collar	Hedging structure that caps upside and floors downside on annual equity returns. Net credit (negative price) means selling the cap pays more than buying the put.
Holiday Mechanism	When the investment account drops below the entry threshold, interest payments to the funder are deferred. Repaid when account recovers above exit threshold.
Pareto Front	Set of configurations where no alternative exists that is both lower risk AND higher return. Also called the efficient frontier.
Profit Share	Percentage of positive surplus extracted every 5 years. FutureProof's primary revenue source.
P1, P10, P99	1st, 10th, 99th percentiles of the surplus distribution across 50,000 simulated paths.